MA1503 Linear Algebra: Final Exam Review

Comprehensive Analysis of 2020–2024 Exam Trends & Solutions

April 20, 2026

Agenda

1. Part I: Exam Trend Analysis (2020–2024)
2. Part II: Short Answers & Basic Concepts (Q1)
3. Part III: Logic Proofs (Q2 - Q5)
4. Part IV: Eigenvalue & Diagonalization & Orthogonality (Q6 - Q12)
5. Part V: Least Squares & SVD (Q13 - Q15)
6. Conclusion
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Part I: 5-Year Topics Comparison Table

Q#	Final 24-25	Final 23-24	Final 22-23	Final 21-22	Final 20-21
1	RREF, Invertible, Basis	Linear Transf.	Linear System Sol.	Invertible, det, Inverse	RREF, Rank
2	det	Subspace	Eigenvalue (prove)	Similar, Diagonal	Inverse Matrix
3	Eigenvalue (prove)	det, Invertible	Orthogonal	Complex Eigenvalue	det
4	Stochastic Matrix	Diagonal	Least-Square	Linear Transf.	Eigen., Orth., Diag.
5	Orthogonal	Eigenvalue (prove)	Diagonal	Orthogonal	Orth., Least Square
6	Least-Square	Least Square	Vector Space (prove)	Least Square	Prove (Subspace)
7	Diagonal	Orthogonal	Singular Value	Eigenvalue	Prove (Rank)
8	Singular Value	–	–	Singular Value	Prove (Eigenvalue)

- Red text: High-Frequency Topics. Yellow background : Proof-based questions.

Part I: Macro Trend Analysis

- **Guaranteed Topics (100% Frequency)**:
  - **Det**: foundational topic in linear algebra.
  - **Eigenvalue**: Final **proof-heavy** topics.
  - **Diagonalizability**:Foundation of Orthogonal Singular Value Decomposition.
  - **Orthogonality**: The most stable concept across all years.
  - **Least-Squares**: **Most Accessible** Fundamental Technique.
  - **Singular Value Decomposition**:Most comprehensive topic.
- **The Evolution**: Increased emphasis on **conceptual mastery and proof-based** problems.
- **Foundational knowledge is paramount**: Most questions merely test core mastery, making a thorough review of lecture notes **absolutely necessary**.

Part I: Difficulty Distribution Strategy

Q1 - Q3, Q7, Q9, Q11 - Q13 **The Safety Zone**: Definitions, basis menthods and determinants. **High accuracy required**.

Q4 - Q6, Q8, Q10 **The Logic Zone**: Subspace proofs, eigenvalue and Matrix Analysis. Requires **rigorous derivation and logical analysis** skills.

Q14 - Q15 **The Calculation Zone**:SVD.Require **heavy computational load** proficiency in calculation and **meticulous verification**.

Part II: Question 1 - Definitions and Foundations

Key Solutions:

- (3) Subspace: Zero, Addition, Scalar Mult.
- (7) Least-Squares: $A^T Ax = A^T b$.
- (9) Singular Values: $\sigma_i = \sqrt{\lambda_i(A^T A)}$.
- (10) Optional (Out of Scope): While not examinable, this problem offers valuable insight into SVD and will not be discussed here.

Difficulty: Easy but important
Analysis:
For (6), remember the **parallelogram** area.
For (8), Spectral theorem for symmetric matrices: The following are equivalent.
(a) There exists an orthonormal A-eigenbasis.
(b) A is orthogonally diagonalizable.
(c) A is symmetric.

Part III: Question 2 & 3 - Eigenvalue Logic Proof

Q2(1): Invertibility vs. Eigenvalues

- **Analysis**: If $\lambda = 0 \implies Av = 0 \implies$ dependent columns $\implies$ not invertible.

Q2(2) and Q3(2): Matrix Polynomials

- **Analysis**:Derivation and Analysis Based on the definition $A\mathrm{v} = \lambda\mathrm{v}$, we can derive the following: Matrix Powers (A^k)

$$A^k\mathrm{v} = \lambda^k\mathrm{v}$$

(Derivation logic: $A^2\mathrm{v} = A(A\mathrm{v}) = A(\lambda\mathrm{v}) = \lambda(A\mathrm{v}) = \lambda^2\mathrm{v}$, and so on.)
Scalar Multiples of the Identity (mI)

$$(mI)\mathrm{v} = m\mathrm{v}$$

Since $A^k\mathrm{v} = \lambda^k\mathrm{v}$ and $(mI)\mathrm{v} = m\mathrm{v}$, we have $(\mathrm{A}^k + mI)\mathrm{v} = (\lambda^k + m)\mathrm{v}$.
Thus, $\lambda^k + m$ is an eigenvalue of $A^k + mI$.
Difficulty: Easy

Part III: Question 4 & 5 - Subspaces and Contradictions

Q4: Even Polynomials E_3

- **Analysis**: A general polynomial in P_3 is $p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$. For $p(x) = p(-x)$, we must have $a_1x + a_3x^3 = -a_1x - a_3x^3$ for all x, which implies $a_1 = a_3 = 0$. This is the key step in solving the problem.

Difficulty: Medium

Q5: Contradiction Proof $A = CB$

- **Logic**: **Logic:** Zero row in $B' \implies B'k = 0$ (for $k \neq 0$) $\implies CB'k = 0 \implies A'k = 0$. However, $A'k = 0$ implies only the trivial solution, then $k = 0$. This leads to a contradiction.

Difficulty: Hard (Proof-heavy)

Part IV: Question 6 & 7 - Complex Pairs & Diagonalizability

Q6: Complex Pairs

- Real matrices $\implies$ eigenvalues appear as $1 \pm i$.
- A cubic polynomial with roots x_1, x_2, x_3 can be written in the factored form $f(x) = c(x - x_1)(x - x_2)(x - x_3)$.

Q7: Diagonalizability

- An $n \times n$ matrix is diagonalizable if it possesses n linearly independent eigenvectors (regardless of the number of distinct eigenvalues). Even in the presence of repeated eigenvalues, diagonalizability is guaranteed provided that the geometric multiplicity equals the algebraic multiplicity for each eigenvalue.
- A matrix A is non-invertible $\iff$ 0 is an eigenvalue of A.
  Proof: 0 is an eigenvalue $\iff \exists \mathrm{v} \neq 0, A\mathrm{v} = 0 \cdot \mathrm{v} = 0$. This implies $A\mathrm{v} = 0$ has a non-trivial solution, so the columns of A are linearly dependent, $\det(A) = 0$, and A is non-invertible (singular). **Difficulty**: Medium

Part IV: Question 8 & 10 - Limits and Orthogonal Complements

Q8: Long-term behavior/ Discrete Dynamical Systems $x_{k+1} = Ax_k$

- **Analysis**: Component with $|\lambda| < 1$ vanishes as $k \to \infty$.

Q10: Symmetric Matrices

- **Analysis**: Eigenspaces are orthogonal complements.
- For symmetric matrices, the algebraic multiplicity (AM) always equals the geometric multiplicity (GM) for every eigenvalue. This means you do not need to verify whether $GM = AM$; the matrix is guaranteed to be diagonalizable.

Difficulty: Medium

Part IV: Question 11 & 12 - Projections and Gram-Schmidt

Q11: Projection Matrix

- $\text{proj}_V(x) = \frac{x \cdot v}{v \cdot v} v + \frac{x \cdot u}{u \cdot u} u, \quad$ where $v \perp u$.

Q12: Best Approximation

- When using basis vectors for projection, it is imperative to first verify that they are mutually orthogonal.

Difficulty: Easy

Part V: Question 13 - Least-Squares Truths

- (a) **False**: The solution to $A^T A\mathrm{x} = A^T \mathrm{b}$ is the least-squares solution. Only when the original system $A\mathrm{x} = \mathrm{b}$ is consistent (i.e., b lies in the column space of A) does the least-squares solution satisfy $A\mathrm{x} = \mathrm{b}$. If $A\mathrm{x} = \mathrm{b}$ is inconsistent, $A\hat{\mathrm{x}}$ is merely the projection of b onto the column space and does not equal b.
- (c) **False**: The normal equation $A^T A\mathrm{x} = A^T \mathrm{b}$ always has a solution. This is because the vector $A^T \mathrm{b}$ always lies in the column space of $A^T A$. Consequently, regardless of whether A is full rank or the position of b, there exists at least one least-squares solution $\hat{\mathrm{x}}$.
- (e) **True**: The normal equation $A^T A\mathrm{x} = A^T \mathrm{b}$ has a unique solution if and only if the columns of A are linearly independent. If the columns of A are linearly independent, then for any equation $A\mathrm{x} = \mathrm{b}$: if b lies in the column space of A, there is a unique solution; if b is not in the column space, there is no solution. Both cases satisfy the condition of having **at most one solution**.

Difficulty: Medium

Part V: Question 14 & 15 - The SVD Mastery

Q14 (2): Geometry of SVD

- **Visualization**: Optional (Out of Scope): While not examinable, this problem offers valuable insight into SVD and will not be discussed here.

Q15: Very Hard

- Although this question is challenging, it is highly valuable for understanding SVD.
- A fundamental property of an orthogonal matrix Q is that $Q^T Q = I$ (or equivalently, $QQ^T = I$). Use this property to find a nonzero vector in $\mathrm{Nul}(A)$.
- Determine Σ: Given $\sigma_{\max} = 22$ and $\sigma_{\min} = 20$, we directly obtain $\Sigma = \begin{bmatrix} 22 & 0 \\ 0 & 20 \end{bmatrix}$.
- Given $A \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix} = \begin{bmatrix} 0 \\ 20 \end{bmatrix}$ Since the output length is 20, which corresponds to the minor axis (the smaller singular value), the input vector must be the second column v_2 of V, and the output direction must be the second column u_2 of U.

Difficulty: Hard, Q15 is the hardest question of these 15 questions

Final Exam Strategy

Mastering the MA1503 Final

1. **The Spectral Theorem**: Real symmetric matrices are always orthogonally diagonalizable; their eigenspaces are mutually orthogonal, guaranteeing the existence of n mutually perpendicular unit eigenvectors.
2. **Gram-Schmidt First**: Essential for projections.
3. **Verification for Least Squares**: Although multivariable integration (MA2508) is prohibited in formal answers, you may use it for verification purposes on scratch paper.
4. **Eigenvalue Logic**:$(\mathrm{A}^k + mI)\mathrm{v} = (\lambda^k + m)\mathrm{v}$.

4. **Rank**:Rank-Nullity Theorem: $\operatorname{rank}(A) + \dim(\operatorname{Nul} A) = n$.
   SVD Rank Property: The number of non-zero singular values equals $\operatorname{rank}(A)$.
   Invertibility Equivalences: Full rank $\iff$ $\det(A) \neq 0$ $\iff$ $A\mathrm{x} = 0$ has only the trivial solution $\iff$ 0 is not an eigenvalue

Contact Information

Reach out if you have any questions:

Good luck with your MA1503 Final Exam!